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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/738,626	12/20/2010	Jae Kwan Hwang	86152(307161)	3063
90238	7590	03/27/2014		
Edwards Wildman Palmer LLP P.O. Box 55874 Boston, MA 02205			EXAMINER KAROL, JODY LYNN	
			ART UNIT 1627	PAPER NUMBER
			NOTIFICATION DATE 03/27/2014	DELIVERY MODE ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 12/738,626	Applicant(s) HWANG ET AL.	
	Examiner JODY KAROL	Art Unit 1627	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 3/11/2014.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-5 and 9-12 is/are pending in the application.
5a) Of the above claim(s) 1-5 and 10-12 is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 9 is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☐ The drawing(s) filed on _____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- 1) ☐ Notice of References Cited (PTO-892)
- 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date 3/11/2014.
- 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____.
- 4) ☐ Other: _____.

DETAILED ACTION

Continued Examination Under 37 CFR 1.114

1. A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 3/11/2014 has been entered.

Receipt is acknowledged of applicant's Amendment/Remarks filed 3/11/2014. Claim 9 has been amended. Claims 6-8 are cancelled. Claims 1-5 and 10-15 remain withdrawn as pertaining to the non-elected invention. Claim 1-5 and 8-12 are pending. Claim 9 is currently under consideration.

Information Disclosure Statement

2. The information disclosure statement (IDS) filed on 2/20/2004 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement has been considered.

Response to Arguments

3. Applicant's arguments filed 3/11/2014 have been fully considered but they are not persuasive.

Applicant argues that the BhavamiSra Bhavaprakasa reference does not specifically disclose the macelignan and does not disclose that the macelignan compound is useful for inhibiting melanin production. In response to applicant's arguments against the references individually, one cannot show nonobviousness by attacking references individually where the rejections are based on combinations of references. See *In re Keller*, 642 F.2d 413, 208 USPQ 871 (CCPA 1981); *In re Merck & Co.*, 800 F.2d 1091, 231 USPQ 375 (Fed. Cir. 1986). As taught by Hwang et al., macelignan is clearly present in *Myristica fragrans* (i.e. nutmeg) which is used in the formulation taught by BhavamiSra Bhavaprakasa reference.

Applicant further argues that although the extracts of *Myristica fragrans* have been used in various food and pharmaceutical purposes, it has been found that the extract of *Myristica fragrans* has more than a dozen pharmaceutical effects, wherein various chemical components of extract have been revealed, and wherein some of these are considered to be responsible for the pharmacological effects of *Myristica fragrans*. Applicant argues that arduous scientific studies are still required to isolate and determine which compounds (either solely, or in combination with other phytochemicals) are responsible for the effects of *Myristica fragrans*. Applicant argues that there are relatively few scientific studies disclosing specific phytochemicals responsible for a certain pharmacological effect of *Myristica fragrans*, and that it is probably that further pharmacological effects and/or phytochemicals have yet to have been discovered. Applicant argues that considering the inherent complexity and unpredictability of chemical interactions and resulting effects among various chemical components

included within a plant extract, it would have been difficult for a person of ordinary skill in the art to isolate and determine which single compound or combination of many different components as included in the extract of *Myristica fragrans* produce a specific pharmacological effect, even in the extract of *Myristica fragrans* has been traditionally reported as "so-called folk medicine in India" that it could be useful for treating excess melanin-related skin conditions. The Examiner respectfully disagrees. The BhavamiSra Bhavaprakasa reference clearly teaches nutmeg formulation is useful in treating excess melanin-related conditions such as freckles. It would have been obvious to one of ordinary skill in the art to determine and isolate the active components in *Myristica fragrans* for inhibiting melanin production, a single pharmaceutical use. One of ordinary skill in the art would have been motivated to determine and isolate the active components for inhibiting melanin production with a reasonable expectation of success being the useful parts of *Myristica fragrans* are known to be useful in the Jatiphalalepah formulation which is known for treating conditions related to excess melanin (i.e. freckles), and which contains macelignan as taught by Hwang et al.

Applicants further argue that in response to the June 24, 2013 Non-Final Office Action, Applicants asserted (and submitted 3 documents) that even structurally similar lignan compounds might possess significantly different and unexpected therapeutic effects (included opposite effects) from that of the macelignan compound according to the pending claims. In response it is respectfully submitted that there is a reasonable expectation that the macelignan, a lignan compound found in seed, fruits or arils of *Myristica fragrans* as taught by Hwang et al., would be useful in the treatment of

freckles and cholasma/melasma/melanoderma (i.e. abnormal darkening of the skin caused by excess melanin) because the Jatiphalalepah formulation for the treatment of said conditions, contains the useful parts of *Myristica fragrans*. Further, one of ordinary skill in the art would have been motivated to determine and isolate the active components for melanin inhibition in view of the documents submitted (Matsuda et al., Hirata et al., and JP 2005-289838A) because it would be desirable formulation a composition for melanin inhibition without components that have opposite of the desired effect.

Applicants argue that the pharmacological activity of the macelignan compound as described in Hwang et al. is significantly different from the compound of the instant claims. Applicants further argue a person of ordinary skill in the art would not have assumed that the macelignan represented by the Chemical Formula 1 would be useful for inhibiting melanin production in a subject in need thereof. In response it is respectfully submitted that the macelignan compound taught by Hwang et al. is the same compound as instantly claimed and thus, is expected to have the same properties as instantly claimed. "Products of identical chemical composition can not have mutually exclusive properties." A chemical composition and its properties are inseparable. Therefore, if the prior art teaches the identical chemical structure, the properties the applicant discloses and/or claims are necessarily present. *In re Spada*, 911 F.2d 705, 709, 15 USPQ2d 1655, 1658 (Fed. Cir. 1990). Further, the macelignan is a known compound present in *Myristica fragrans*, wherein the useful parts of *Myristica fragrans* are known to be useful in the Jatiphalalepah formulation for treating conditions related

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to excess melanin (i.e. freckles). Thus, it would have been obvious to for one of ordinary skill in the art to try treating conditions related to excess melanin (i.e. freckles) with the macelignan compound. Further, the determination and isolation of active components in a known formulation for the same purpose is considered to be within the purview of the ordinary artisan.

Thus, for these reasons, Applicant's arguments are found unpersuasive. Said rejection is maintained.

MAINTAINED REJECTIONS

4. The following rejections have been maintained from the previous Office Action dated 9/11/2013 but have been amended to account for Applicant's amendment to claim 9:

Claim Rejections - 35 USC § 103

5. The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103(a) are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.
4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims under 35 U.S.C. 103(a), the examiner presumes that the subject matter of the various claims was commonly owned at the time any inventions covered therein were made absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and invention dates of each claim that was not commonly owned at the time a later invention was made in order for the examiner to consider the applicability of 35 U.S.C. 103(c) and potential 35 U.S.C. 102(e), (f) or (g) prior art under 35 U.S.C. 103(a).

6. Claim 9 is rejected under 35 U.S.C. 103(a) as being unpatentable over BhavamiSra; Bhavaprakasa (Edited & translated by Brahmasankara Misra, Part-II: Chaukhamba Sanskrit Sansthan, Varanasi, Edn. 7th, 2000. [Time of origin 16th century], pg 589; Formulation Id: RG/2838A; Formulation Name: Jatiphalalepah; Publication (Prior art): pg No.03; English Translation including Terminology Conversion (TKDL Extracts): pg No.04-05 – cited on IDS) in view of Hwang et al. (KR 1020050073027 A). It is noted that US 2009/0192217 A1 is used as the English language equivalent of the KR publication for citation purposes.

The formulation Jatiphalalepah is a therapeutic single compound formulation consisting of the useful parts of the nutmeg tree *Myristica fragrans* Houtt, wherein the ingredients are made into a fine power and mixed with liquid to form a paste. The paste

is locally applied, such as to the face, for the treatment of freckles and cholasma/melasma/melanoderma (i.e. abnormal darkening of the skin caused by excess melanin).

The formulation does not specifically teach that a lignan-type compound of Chemical Formula 1 (macelignan) is used to inhibit melanin production.

However, Hwang et al. teach lignan compound such as macelignan (Chemical Formula 1) are found in the seeds, fruits, or arils of *Myristica fragrans* wherein said lignans can be obtained from pressing the seeds to extract the oil (see pages 2 and 3, sections [0019]-[0024]).

It would have been obvious to one of ordinary skill in the art at the time of the invention to determine and separate out the active components, i.e. the macelignan as taught by Hwang et al to be present in *Myristica fragrans* (i.e. nutmeg), from the useful parts of nutmeg used for treating freckles or cholasma/melasma/melanoderma as taught by the Jatiphalalepah formulation. One of ordinary skill in the art would have been motivated to determine and separate out the active components, i.e. the macelignan as taught by Hwang et al to be present in *Myristica fragrans* (i.e. nutmeg), from the useful parts of nutmeg used for treating freckles or cholasma/melasma/melanoderma as taught by the Jatiphalalepah formulation in order to formulate a more potent composition for use in treating freckles or cholasma/melasma/melanoderma. One of ordinary skill in the art would have had a reasonable expectation of success in determining and separating out the active components, i.e. the macelignan as taught by Hwang et al to be present in *Myristica*

fragrans (i.e. nutmeg), from the useful parts of nutmeg used for treating freckles or cholasma/melasma/melanoderma as taught by the Jatiphalalepah formulation because the isolation of active components from natural product extracts is *prima facie* obvious and a matter of routine experimentation.

Further, it would have been obvious to one of ordinary skill in the art at the time of the invention that in applying the active component isolated from the Jatiphalalepah formulation to the face of a person with freckles or cholasma/melasma/melanoderma, melanin production would be inhibited. Increased melanin production is considered to be a mechanism of action for freckles or cholasma/melasma/melanoderma. The mechanism of action does not have a bearing on the patentability of the invention if the invention was already known or obvious. Mere recognition of latent properties in the prior art does not render nonobvious an otherwise known invention. *In re Wiseman*, 201 USPQ 658 (CCPA 1979).

Thus, the invention as a whole would have been *prima facie* obvious to one of ordinary skill in the art at the time of the invention.

NEW REJECTIONS

7. In view of Applicant's amendment to claim 9 and upon further consideration, the following rejection is newly applied:

Claim Rejections - 35 USC § 101

8. 35 U.S.C. 101 reads as follows:

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Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

9. Claim 9 is rejected under 35 U.S.C. 101 because the claimed invention is not directed to patent eligible subject matter. Based upon an analysis with respect to the claim as a whole, claim(s) 9 do not recite something significantly different than a judicial exception. The rationale for this determination is explained below:

Claim 9 is directed to a method of using isolated macelignan. As evidenced by Hwang et al., macelignan is not-naturally occurring because it occurs naturally in *Myristica fragrans* (i.e. nutmeg). There is no structural difference between the macelignan of the instant claims and the macelignan of nutmeg because removal or "isolation" of the macelignan from nutmeg did not change its structure. Further, claim 9 recites only a general method step of administering the compound, which is a conventional and routine method step that amounts nothing more to a field of use.

Conclusion

No claims are allowed.

Correspondence

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Jody L. Karol whose telephone number is (571)270-3283. The examiner can normally be reached on 8:30 am - 5:00 pm Mon-Fri EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Sreeni Padmanabhan can be reached on (571) 272-0629. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

/Jody L. Karol/

Examiner, Art Unit 1627

/SREENI PADMANABHAN/
Supervisory Patent Examiner, Art Unit 1627